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MTS 210702 - Principles of Mechatronics

Group Project:

Real-Time Mechanical Vibration Analysis Prototype for Maintenance

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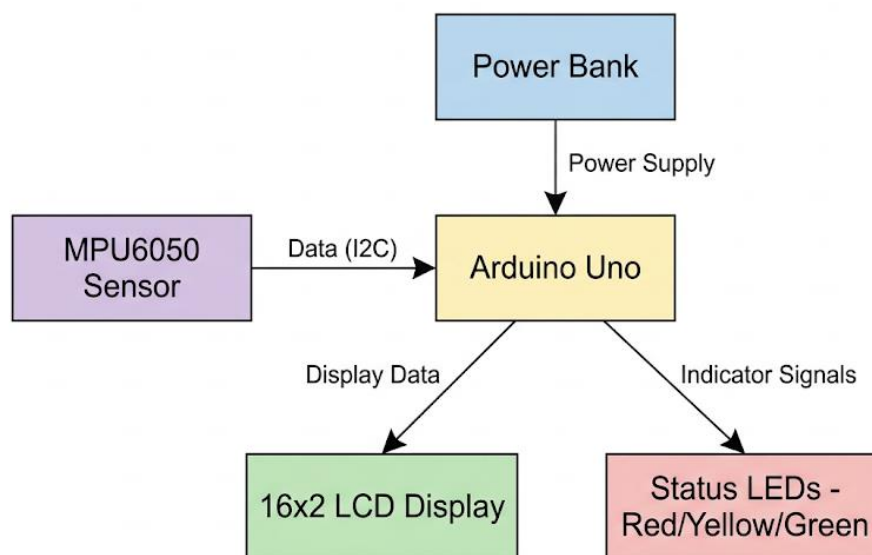
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1. Introduction

In modern industrial operations, the shift from reactive to proactive maintenance is essential for maximizing efficiency. This prototype implements a **Condition-Based Maintenance (CBM)** strategy, focusing on the current health state of rotating equipment (pumps, fans, motors). Utilizing an MPU6050 MEMS-based accelerometer, the system performs real-time, non-invasive assessment of machine oscillations. The primary objective is to detect faults such as mass imbalance, shaft misalignment, and structural looseness before they escalate into catastrophic failures.

2. System Architecture

Block Diagram



3. Control Explanation

The prototype employs **Condition-Based Maintenance (CBM)** logic through **Threshold-Based Amplitude Analysis**. It operates strictly in the time domain, focusing on peak-to-peak acceleration magnitude. This method is the established industrial entry point for **Fault Detection**, providing a localized, instantaneous assessment of a machine's current health state. By processing data at the "Edge" (locally on the Arduino), the system provides immediate, zero-latency diagnostic feedback to the operator.

How it Works

1. **Normalization:** The controller calculates the total acceleration magnitude:

$$\text{Magnitude} = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

2. **Calibration:** To isolate mechanical vibration, the system subtracts the constant effect of Earth's gravity ($1g \sim 9.81\text{m/s}^2$)
3. **Classification:** Following the **ISO 10816-3** guidelines for Class I machinery, the vibration is classified into three zones:

ISO 10816 / 20816 Overview

ISO 10816 (now superseded by ISO 20816) classifies vibration severity based on broadband velocity in mm/s RMS, measured on the bearing housing of rotating machines. The standard defines four evaluation zones:

- **Zone A** — Vibration of newly commissioned or reconditioned machines
- **Zone B** — Acceptable for unrestricted long-term operation
- **Zone C** — Not suitable for long-term continuous operation; remedial action required
- **Zone D** — Dangerous vibration levels; damage could occur; immediate action needed

Machine Classes

CLASS	DESCRIPTION	EXAMPLES
I	Small machines ≤ 15 kW	Small pumps, fans, electric motors
II	Medium machines 15–75 kW (or ≤ 300 kW on special foundations)	Medium motors, pumps, generators
III	Large machines > 300 kW on rigid/heavy foundations	Large motors, centrifugal compressors
IV	Large machines > 300 kW on flexible/lightweight foundations	Turbogenerators, gas turbines, large fans

Zone Boundary Values (mm/s RMS)

BOUNDARY	CLASS I	CLASS II	CLASS III	CLASS IV
A/B	0.71	1.12	1.80	2.80
B/C	1.80	2.80	4.50	7.10
C/D	4.50	7.10	11.2	18.0

Reference: <https://vibromera.eu/calculators/vibration-severity-chart/>

Threshold Selection and System Response

We have established vibration thresholds of **2.0g** and **5.0g** to categorize machine health based on **ISO 10816-3** severity standards for Class I machinery. These values were chosen because acceleration (g) is highly sensitive to high-frequency faults like early bearing wear and mechanical looseness.

Normal Condition (< 2.0g): Represents a healthy state; the system illuminates a **Green LED**, indicating the machine is within the "Satisfactory" zone.

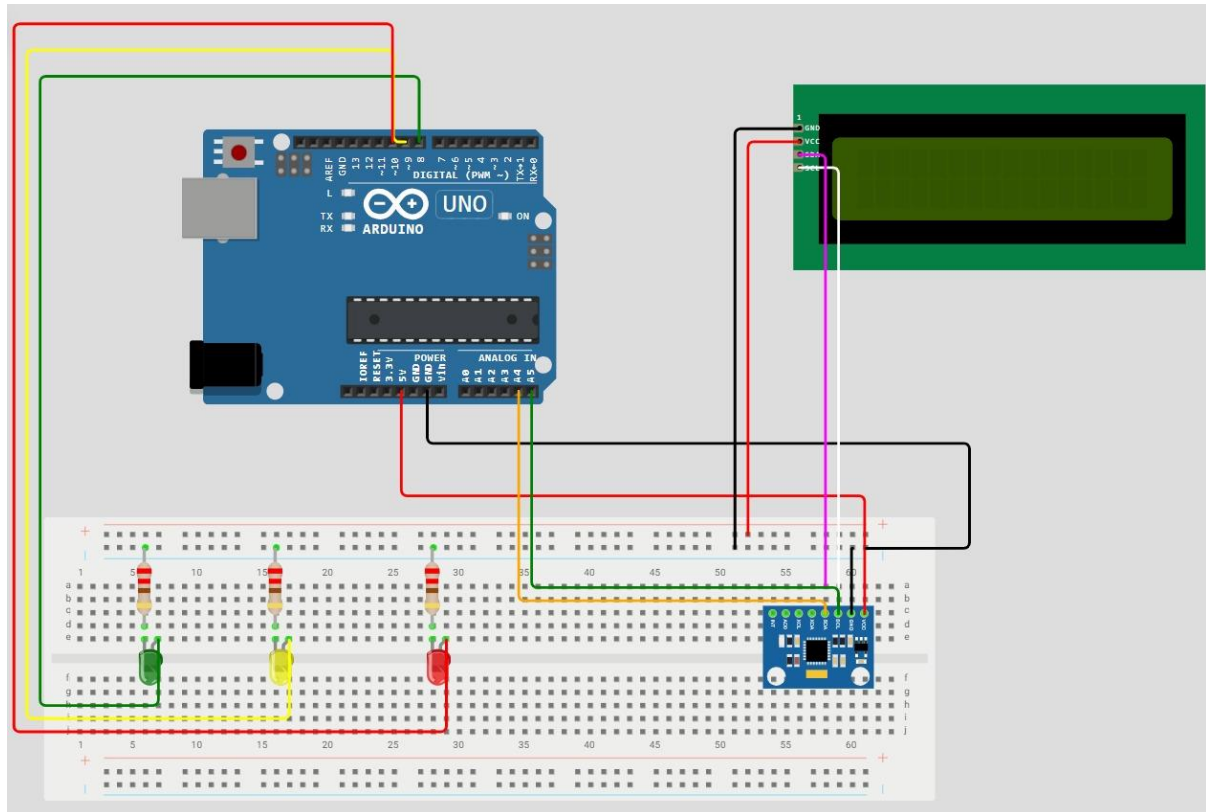
Warning Condition (2.0g – 5.0g): Triggers a **Yellow LED** to alert the operator of developing faults such as **mass imbalance** or **shaft misalignment**.

Critical Alert (> 5.0g): Activates a **Red LED**, signaling an "Unacceptable" condition that requires an immediate shutdown to prevent catastrophic failure.

To ensure continuous monitoring, the real-time G-force magnitude is constantly displayed on the **16x2 LCD**, providing the operator with precise numerical data for proactive decision-making

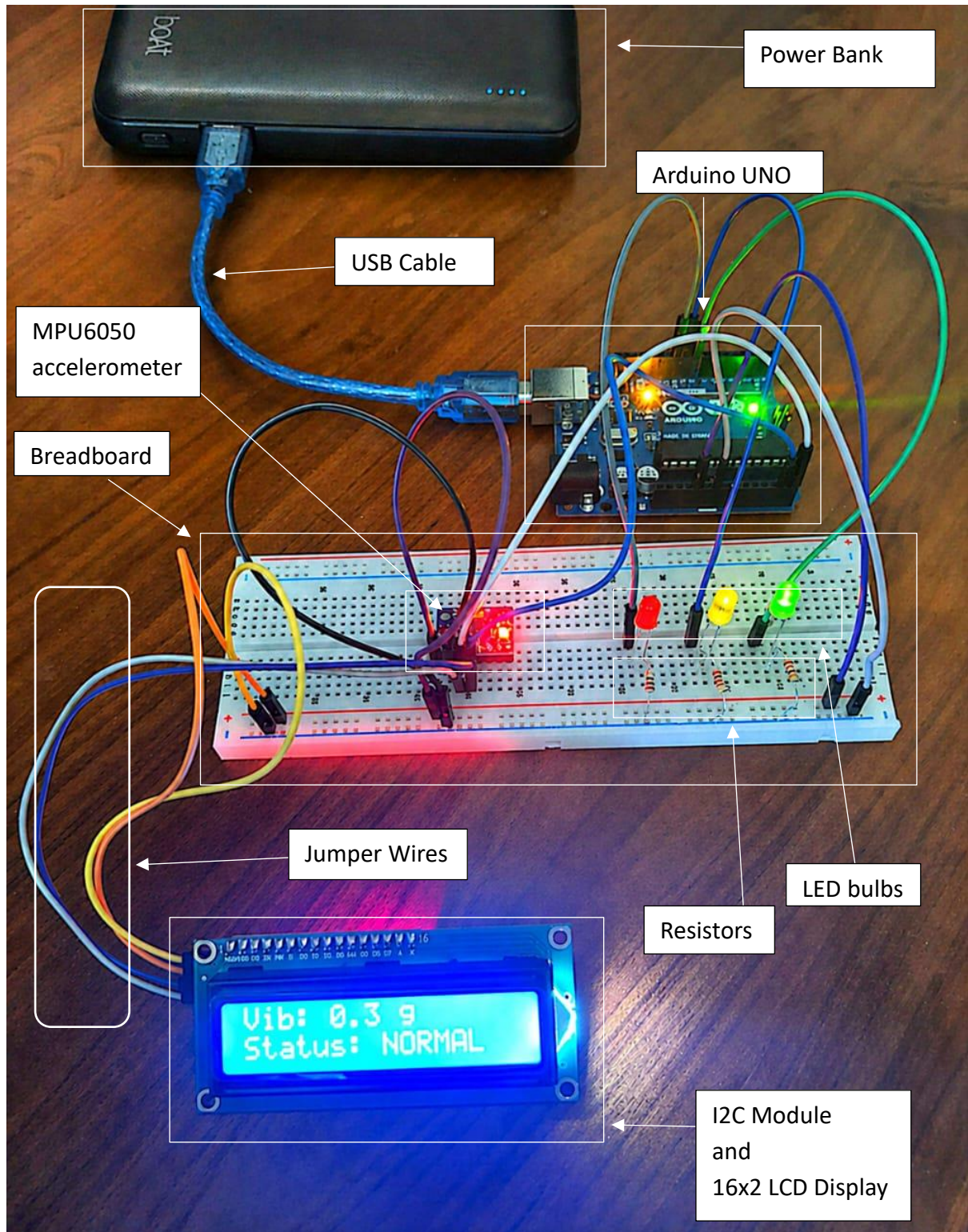
Hardware Evidence

Circuit Diagram

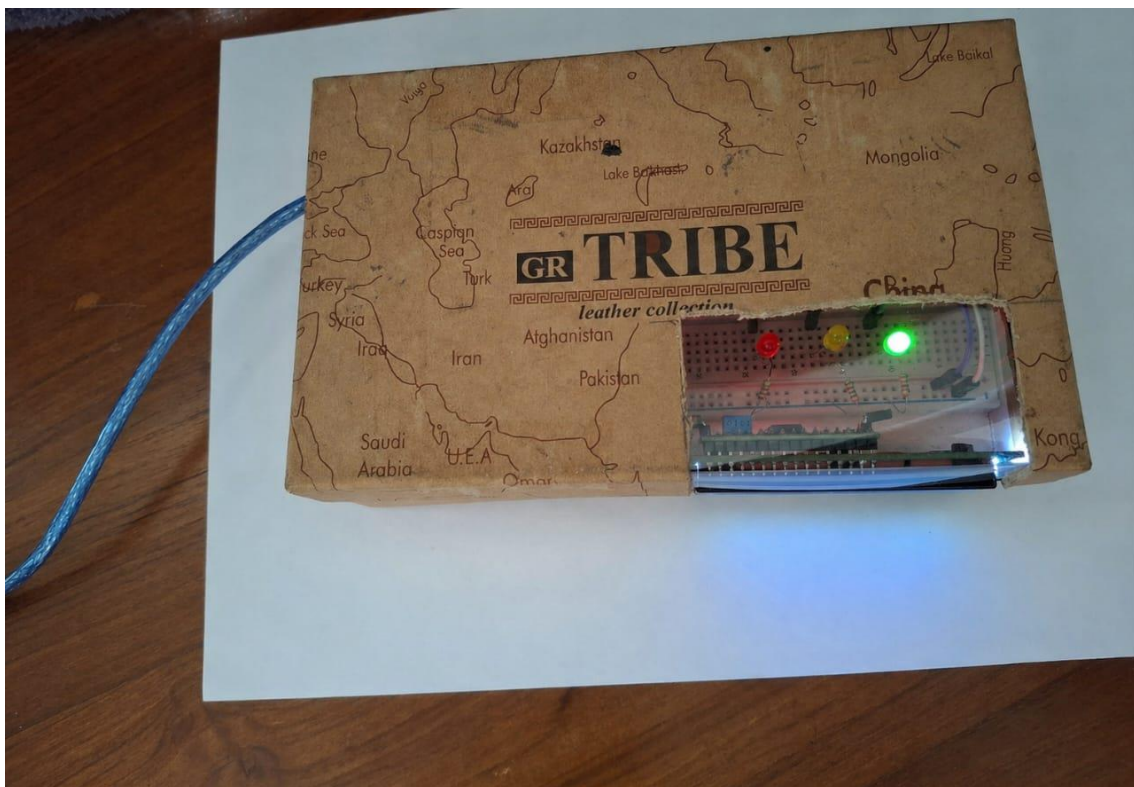
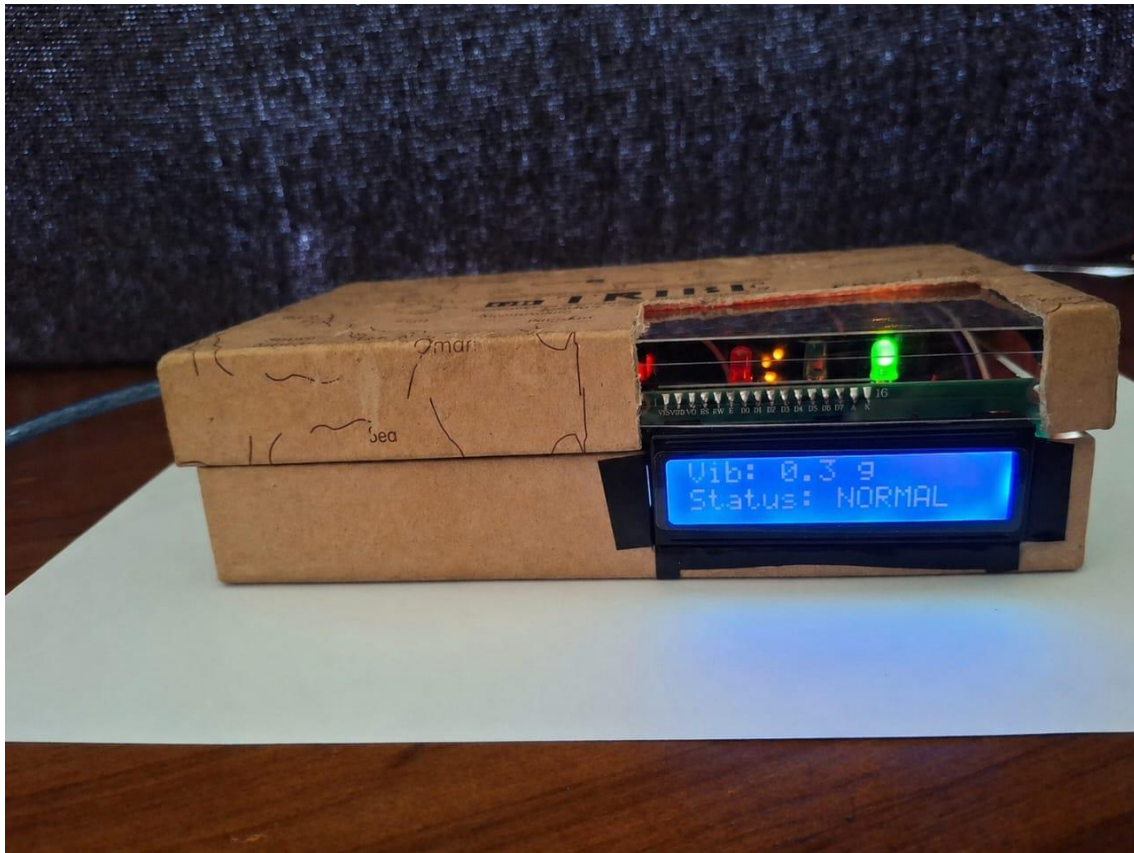


Reference: <https://wokwi.com/projects/461646299031681025>

Labeled Picture of the Wiring



Final Assembly



Industry Use-Case

Comparison with Leading Industrial Solutions

To understand the "Industry Gap" this project addresses, it is essential to compare our prototype with existing market leaders in the vibration monitoring sector. Currently, industries rely on high-fidelity, wireless nodes from globally recognized brands. However, as shown in the table below, these systems come with significant costs, ranging from **\$500 to over \$5,000 per node**, excluding additional gateway and subscription fees.

Company & Product	Approx. Price	Key Features	The Gap vs. Our Solution
Lord Microstrain	\$500 - \$1,000	Rugged, 800m range	High hardware cost per node.
enDAQ W-Series	\$3,000 - \$4,000	20 kHz sampling, 16GB storage	Prohibitive price for small scale.
SKF Enlight IMx-1	\$5,000+ (System)	Full factory integration	Requires proprietary gateways.
Our Prototype	Under \$10	Real-time Edge Alerts	Affordable for SMEs.

Reference: <https://blog.endaq.com/top-accurate-wireless-vibration-monitoring-systems>

The Industry Gap & Our Solution

The table highlights a major barrier for Small and Medium Enterprises (SMEs): **Cost and Complexity**. While systems like Bently Nevada or enDAQ offer extreme precision, they are often financially out of reach for smaller workshops.

Our prototype fills this gap by providing:

1. **Affordability:** Built at a fraction of the cost (\$20 vs \$1,000+).
2. **Edge Processing:** Our system analyzes data locally on the Arduino, providing instant alerts via LCD/LED without needing a \$2,000 gateway.
3. **Accessibility:** It simplifies vibration monitoring so that any technician can understand the machine health instantly (Red/Yellow/Green), without needing complex data analysis software.